

InnovaHack-2026

ACETRACK

Personalized AI Study Coach

Team: Chakra

Members:

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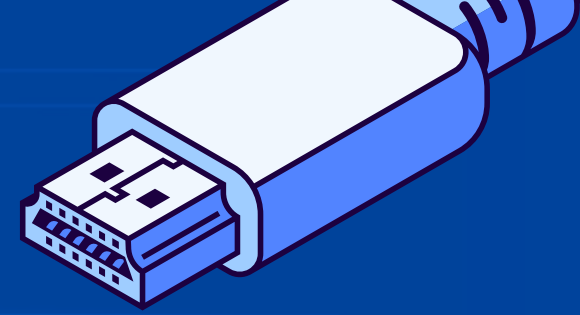
 Indraprastha Institute of Information
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AI that doesn't just test students—it understands how they learn and adapts to help them improve.



PROBLEM STATEMENT 2

PERSONALIZED AI STUDY INTERVIEW COACH



Students struggle because:

- Static question banks
- No personalized feedback
- No adaptive study planning
- Difficult to identify weak concepts
- No intelligent resource recommendations

Our Solution - AceTrack

- AI-generated mock tests
- Intelligent answer evaluation
- Adaptive study planner
- Feedback and source recommendation
- Progress tracking & gamification

Topic Selection



AI Quiz Generation



Attempt Test



AI Evaluation



Feedback & Resources

AI QUIZ GENERATOR

Students can create completely customized mock tests by selecting:

- Subjects
- Topics
- Question formats
- Number of questions

With many question types as shown in the image →

The screenshot shows the 'Create a New Mock Test' page in the AceTrack application. At the top, there's a header with the AceTrack logo and a 'Mock Tests' button. The main heading is 'Create a New Mock Test' with a subtext 'Generate a personalized test based on your syllabus.' Below this, step 1 'Select Syllabus' has a yellow warning box stating 'No syllabus found. Please go to [Syllabus Settings](#) to upload one first.' Step 2 'Select Topics by Subject (1 to 15 max)' shows three subject cards: Physics, Chemistry, and Maths. Each card has a '0 selected' indicator and a dropdown menu to 'Choose [Subject] Topic'. A progress indicator at the top right of this section shows 'Selected 0 / 15 topics'.

This screenshot displays the question selection interface in the AceTrack application. At the top, there's a header with the AceTrack logo and a 'Mock Tests' button. Below the header, there are radio buttons for 'PDF' (selected) and 'DOCX'. The main section is titled '3. Select Question Counts (1 to 15 max total)' with a progress indicator 'Total Questions: 0 / 15 max'. The interface lists various question types, each with a description and a selection box showing '0':

- MTF**: Match the Following questions.
- 2S**: Two-statement reasoning questions.
- 3S**: Three-statement analysis questions.
- 4S**: Four-statement analysis questions.
- 5S**: Five-statement, highly analytical questions.
- SL**: Single-liner scenario-based questions.
- AR**: Assertion and Reasoning questions.
- CS**: Case study comprehension questions.
- CH**: Chronological ordering questions.
- FU**: Fill in the Blanks questions.
- MCQ**: Multiple choice questions.
- NU**: Numerical answer type questions.

At the bottom, there is a large blue button labeled '+ Generate Test'.

AI EVALUATION & PERSONALIZED FEEDBACK

Instead of only showing the score, AceTrack explains performance.

After submission, AI:

- ✓ Evaluates answers
- ✓ Identifies weak concepts
- ✓ Highlights strengths
- ✓ Suggests improvements
- ✓ Recommends reference books

Example Recommendations

- H.C. Verma
- R.D. Sharma
- O.P. Tandon

Students receive actionable feedback rather than just a score.

Mock Test

Score: 7 / 12

AI Feedback

You've done a commendable job scoring 7 out of 12 on your mock test! It's great to see that you correctly answered questions about the centroid in a triangle and Newton's third law of motion, showing a solid understanding of both geometry and physics fundamentals. However, it seems like you might need to review the Kinetic Theory of Gases, as there was a mix-up in that question. Focusing on this area will help you strengthen your grasp on the behavior of gases, which is a crucial topic in physics.

Recommended Books

- H.C. Verma - Concepts of Physics
- O.P. Tandon - Physical Chemistry

Q1. Which of the following best defines the behavior of gases according to the Kinetic Theory?

A. Gases consist of particles in random motion with no interactions

B. Gases are incompressible

C. Gases have fixed shapes and volumes

D. Gases consist of large particles with strong intermolecular forces

Solution: The Kinetic Theory of Gases states that gases consist of many small particles that are in constant, random motion and interact only through elastic collisions. Hence, Option (1) is the right answer.

Q2. In a triangle, the point where the medians of the triangle meet is called the:

A. Circumcentre

B. Orthocentre

C. Centroid

D. Incentre

Solution: The centroid is the point where the medians of the triangle intersect, and it is also known as the center of mass or barycenter of the triangle. Hence, Option (3) is the right answer.

STUDY PLANNER

1

Students just enter their exam, date of the exam, daily study hours, and topics already covered, and AceTrack builds the rest.

The screenshot shows the 'Generate Study Plan' interface in the AceTrack app. It includes the following fields and options:

- 1. Exam Name:** A dropdown menu with 'JEE' selected.
- 2. Topics Already Done (Optional — excluded from your plan):** Three boxes for 'Physics', 'Chemistry', and 'Maths', each with a '0 done' status and a '-- Select Done' button.
- 3. Exam Date:** A date input field showing '24/09/2026'.
- 4. Daily Study Hours:** A text input field with the value '4'.
- 5. Days Per Week Available:** A dropdown menu showing '6 Days / week (1 rest day)'.
- + Generate My Plan:** A large purple button at the bottom.

2

- Generates day by day schedule
- Prioritizes high weightage topics first avoids low-priority topics
- Rebalance & Full Regen to stay on track

The screenshot shows the 'Study Plan (JEE)' dashboard in the AceTrack app. It includes the following elements:

- Target Exam Date:** 24 August 2026.
- Buttons:** 'Rebalance' and 'Full Regen' buttons.
- Warning:** A yellow box stating '62 lower-priority topic(s) were excluded from this schedule. Due to limited time budget before your exam date, the highest-weightage topics were prioritized. View excluded topics (62)'.
- Overall Progress:** A progress bar showing '0/56 tasks - 0%'.
- Navigation:** Tabs for 'Today', 'Calendar', 'Progress', and 'Left-Out Topics (62)'.
- Calendar:** A calendar for July 2026 with dates 1 through 31. The 26th is highlighted.
- Task List:** A list of tasks for 'Sun, 26 Jul' with a '0/2 done' status. The tasks are:
 - Physics (2h):** Current Electricity Current, drift velocity and mobility; Ohm law; resistance, resistivity and conductivity; I-V characteristics; electrical energy and power; resistor combinations and temperature dependence; cells, Kirchhoff laws, Wheatstone and metre bridges.
 - Chemistry (2h):** Aldehydes, Ketones and Carboxylic Acids Carbonyl-group nature and nucleophilic addition; relative reactivity of aldehydes and ketones; HCN, ammonia derivatives and Grignard additions; oxidation and reduction; α -hydrogen acidity; Aldol, Cannizzaro and Haloform reactions; tests; acidity of carboxylic acids.

PROGRESS TRACKING & GAMIFICATION

1 Daily task completion tracking

Today — Sunday, 26 July 2/2 done

 **Physics** ⌚ 2h
Current Electricity Current, drift velocity and mobility; Ohm law; resistance, resistivity and conductivity; I-V characteristics; electrical energy and power; resistor combinations and temperature dependence; cells; Kirchhoff laws; Wheatstone and metre bridges.

 **Chemistry** ⌚ 2h
Aldehydes, Ketones and Carboxylic Acids Carbonyl group nature and nucleophilic addition; relative reactivity of aldehydes and ketones; HCN, ammonia derivatives and Grignard additions; oxidation and reduction; α -hydrogen acidity; Aldol, Cannizzaro and Haloform reactions; tests; acidity of carboxylic acids.

- Every day's tasks are checked off individually - completing all of them lights up that date on the streak calendar
- Uncompleting a task removes the streak instantly

2 Streak fire appears only when a full day is completed

 **Study Streak** < Jul 26 >

1
days in a row

S	M	T	W	T	F	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
	27	28	29	30	31	

TECHNOLOGY & AI ARCHITECTURE

Frontend: React.js + TypeScript

Tailwind CSS • JWT + Google OAuth • Dynamic State

Backend: FastAPI (Python)

Async REST APIs • Pydantic • Docker Container

Database: PostgreSQL

Hosted on Aiven (managed cloud DB)

AI: OpenAI GPT-4o Pipeline

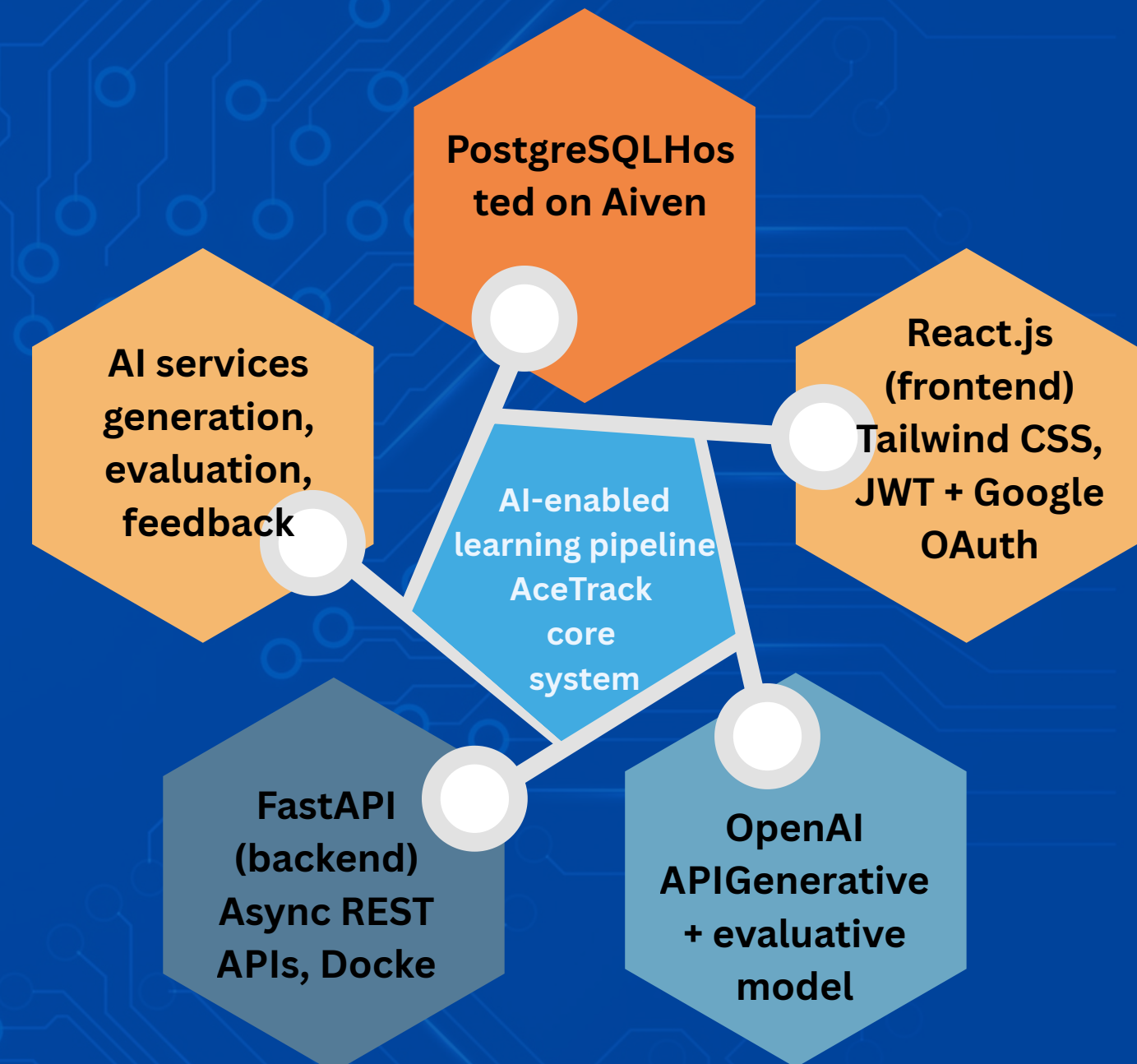
Structured JSON Output • Prompt Engineering

File Storage: Cloudinary

Mock test PDF/DOCX generation & storage

Deployment:

- Render + Docker (Backend)
- Vercel (Frontend)



IMPACT & FUTURE SCOPE

MEASURABLE IMPACT

- 75% Token Cost Reduction: Decouples $O(N)$ deterministic schedule budgeting from GPT-4o calls.
- Automated Assessment Pipeline: Generates 12 question formats directly from raw .xlsx syllabus data.
- Targeted Concept Remediation: Maps missed questions to specific gaps and standard books (H.C. Verma, R.D. Sharma).
- Scalable Async Infrastructure: Containerized FastAPI + PostgreSQL connection pooling on Render/Vercel.

FUTURE TECHNICAL SCOPE

- Open-Model Fine-Tuning (LoRA): Leverage stored MongoDB execution logs to fine-tune Llama-3/DeepSeek for zero-cost offline evaluation.
- Spoken Response & RAG Evaluation: Integrate Whisper API for audio transcription and pgvector RAG over PYQ archives.
- Adaptive Test Engine (IRT): Dynamically scale question difficulty (Factual $\rightarrow$ Analytical) using Item Response Theory.
- Predictive Score Analytics: ML-driven score forecasting based on mastery velocity and topic weightage decay.